Ganji Raghava

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.47
SRI CHAITANYA JUNIOR KALASALA	Class XII (TSBIE)	2023	97.10%
SRI CHAITANYA SCHOOL	Class X (SSC)	2021	100.00%

WORK EXPERIENCE

Backend Developer Intern -- Accion Labs

Dec 2025 - Jan 2026

- Built the backend for a full stack web application in **Node.js**, integrating with a React based frontend to handle natural-language user queries and return responses.
- Designed and implemented backend logic for **input validation, data processing, and routing** between the frontend and application services, ensuring reliable request handling.
- Implemented a **document retrieval feature** that searched across a document set for relevant information and generated contextual answers to natural language user queries.
- Delivered a working end to end prototype within a **4 week internship**, handling **empty and malformed queries** and answering questions across **5 documents simultaneously** through the web interface.

PROJECTS

Knee Arthritis Severity Classification using CNN

Python | *PyTorch* | *NumPy* | *Personal Project*

- Built a **5-class CNN classifier** for knee arthritis severity on **1,650 knee X-rays**, by pixel-variance screening that detected and removed the **12 corrupted scans** before dataset splitting.
- Trained directly on **162×300 X-ray images** using **data augmentation** like **0.5** horizontal flip probability**, ±10°** rotation, and **0.15** brightness & contrast jitter.
- Designed a **4-stage CNN (32 -> 64 -> 128 -> 256 channels)** using **Convolution, Batch Normalization, ReLU activation** and **MaxPooling** layers, with Adaptive Average Pooling producing a **4,096** feature classification head.
- Evaluated **1.25×** and **1.5×** Mild *class weighting* using weighted Cross-Entropy, AdamW. Selected **1.5× Mild *class weighting***, improving **macro-F1** from **0.62->0.66,** achieving **65% accuracy** with **85% Severe-class recall** & **0.81 F1**.

Lossless Data Compression

Course Project, Prof. Shashank Vatedka

- Implemented **LZ77** and **LZ78 algorithms** from scratch in Python, integrating a custom prefix-free integer encoding module to handle dynamic dictionary offsets and reference lengths.
- Built a **Command Line Interface utility** with bit packing algorithms to convert string bitstreams to bytes, saving structured files with embedded **JSON headers**.
- Recorded a **99.2% space reduction** with **LZ77** (shrinking **900 bytes** to **7 bytes**) and an **88.6% reduction** with **LZ78**, while analyzing algorithmic efficiency trends via Matplotlib visualizations.
- Authored a robust **pytest** integration test suite covering edge cases including empty files and high-entropy sequences, ensuring **100% data fidelity** with **0 loss** round-trip restoration.

Credit Card Fraud Detection

Python | *scikit-learn* | *Pandas* | *Group Project*

- Worked on a credit-card fraud detection project involving **~284K transactions across 30 features**, addressing highly imbalanced binary classification with a **0.172% fraud rate**.
- Developed **Logistic Regression** and analyzed its performance against the project's **Neural Network** and **XGBoost** models using **Precision, Recall, F1-score,** and **AUC-PR** on an **80:20 train test split**.
- Selected **AUC-PR** as the primary evaluation metric for the imbalanced problem; removing Time improved AUC PR by **0.06**, with Logistic Regression achieving a **0.76 test AUC PR**.

SKILLS

Programming Languages: Python, C, C++, SQL, JavaScript

Web Technologies: Node.js, HTML, CSS

AI / Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, OpenAI API

Computer Science: Data Structures & Algorithms

RELEVANT COURSES

Data Structures & Algorithms - Pattern Recognition and Machine Learning - Introduction to Modern AI - Probability and Random Variables - Matrix Theory - Convex Optimization

POSITIONS OF RESPONSIBILITY

Student Coordinator -- ML Summer Research Program, IIT Bombay(May - Jul 2025)

- Coordinated **15+** IIT Hyderabad students participating in the **ML Summer Research Program at IIT Bombay** under Prof. Vikram Bansal.
- Managed participant communication and facilitated coordination between students throughout the program.

EXTRACURRICULAR

Silver Medalist -- Institute Hockey League (IHL), IIT Hyderabad (2024)

Interests: Enjoy playing cricket, kabaddi, hockey, and badminton, along with reading in my free time.